King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 231

Original exam date: 7 November 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The area of the surface formed by revolving the graph of $f(x) = x^2$ on the interval $[0, \sqrt{2}]$ about the y-axis is:

(a) $\dfrac{13\pi}{3}$

(b) $\dfrac{10\pi}{3}$

(c) $\dfrac{11\pi}{3}$

(d) $\dfrac{14\pi}{3}$

(e) $\dfrac{16\pi}{3}$

2. If $\int \sec^4 3x \tan^3 3x \, dx = \dfrac{\tan^4 3x}{A} + \dfrac{\tan^6 3x}{B} + C$, then $A + B =$

(a) 26

(b) 32

(c) 30

(d) 28

(e) 6

3. The volume of the solid formed by revolving the region bounded by the graphs of $y = x^2 + 1$, $y = 0$, $x = 0$ and $x = 1$ about the y-axis is: (Hint: The Shell Method is preferable in this question)

(a) $\dfrac{7\pi}{2}$

(b) $\dfrac{\pi}{2}$

(c) $\dfrac{9\pi}{2}$

(d) $\dfrac{5\pi}{2}$

(e) $\dfrac{3\pi}{2}$

4. $\int_0^{\pi/2} \dfrac{d\theta}{1+\sin\theta+\cos\theta} =$

(a) $\ln 3$

(b) $\ln 5$

(c) $2\ln 2$

(d) $\ln 2$

(e) $2\ln 3$

5. The base of a solid is the region bounded by the graphs of $y = x+1$ and $y = x^2-1$. If the cross sections of the solid perpendicular to the x-axis are squares, then the volume of the solid is equal to:

(a) $\int_{-1}^{-2}(x^4+2x^3+3x^2+4x+4)\,dx$

(b) $\int_{-1}^{2}(x^4-2x^3-3x^2+4x+4)\,dx$

(c) $\int_{-1}^{2}(x^4-2x^3-3x^2-4x+4)\,dx$

(d) $\int_{-1}^{2}(x^4+2x^3+3x^2+4x+4)\,dx$

(e) $\int_{-2}^{1}(x^4-2x^3-3x^2+4x+4)\,dx$

6. $\int \dfrac{5x^2+20x+6}{x^3+2x^2+x}\,dx =$

(a) $\ln\left|\dfrac{x^5}{x+1}\right| - \dfrac{9}{x+1} + c$

(b) $\ln\left|\dfrac{x^6}{x+1}\right| - \dfrac{8}{(x+1)^2} + c$

(c) $\ln\left|\dfrac{x^6}{x+1}\right| - \dfrac{9}{x+1} + c$

(d) $6\ln|x| - \dfrac{8}{(x+1)^2} + c$

(e) $\ln\left|\dfrac{x^4}{x+1}\right| - \dfrac{8}{x+1} + c$

7. $\int_0^1 \dfrac{2}{e^{-x}+1}\,dx =$

(a) $2\ln\left(\dfrac{e-1}{2}\right)$

(b) $2\ln\left(\dfrac{e+2}{2}\right)$

(c) $2\ln\left(\dfrac{e+1}{2}\right)$

(d) $\ln\left(\dfrac{e+1}{2}\right)$

(e) $\ln\left(\dfrac{e-1}{2}\right)$

8. $\int \dfrac{x}{\sqrt{4x-x^2}}\,dx =$

(a) $2\arcsin\left(\dfrac{x-2}{2}\right)+\sqrt{4x-x^2}+c$

(b) $2\arccos\left(\dfrac{x-2}{2}\right)-\sqrt{4x-x^2}+c$

(c) $\arccos\left(\dfrac{x-2}{2}\right)-\sqrt{4x-x^2}+c$

(d) $\arcsin\left(\dfrac{x-2}{2}\right)-\sqrt{4x-x^2}+c$

(e) $2\arcsin\left(\dfrac{x-2}{2}\right)-\sqrt{4x-x^2}+c$

9. $\int \dfrac{dx}{\sqrt{x^2-25}} =$

(a) $\ln\left|x-\dfrac{\sqrt{x^2-25}}{5}\right|+c$

(b) $\ln\left|x-\sqrt{x^2-25}\right|+c$

(c) $\ln\left|x+\sqrt{x^2-25}\right|+c$

(d) $\ln\left|\dfrac{x}{5}-\sqrt{x^2-25}\right|+c$

(e) $\ln\left|\dfrac{x}{5}+\sqrt{x^2-25}\right|+c$

10. $\int \dfrac{\ln x}{x^3}\,dx =$

(a) $\dfrac{-\ln x}{2x^2} - \dfrac{1}{4x^2} + c$

(b) $\dfrac{-\ln x}{2x^2} + \dfrac{1}{4x^2} + c$

(c) $\dfrac{-\ln x}{2x^2} - \dfrac{1}{2x^2} + c$

(d) $\dfrac{-\ln x}{x^2} - \dfrac{1}{4x^2} + c$

(e) $\dfrac{\ln x}{2x^2} + \dfrac{1}{4x^2} + c$

11. If $\dfrac{6x}{x^3-8} = \dfrac{A}{x-2} + \dfrac{Bx+C}{x^2+2x+4}$, then $A+B+C =$

(a) 3

(b) 0

(c) -1

(d) 2

(e) 1

12. If $\int \sin\sqrt{x}\,dx = a\sqrt{x}\cos\sqrt{x} + b\sin\sqrt{x} + c$, then $a+b =$

(a) 4

(b) -1

(c) 0

(d) -4

(e) 1

13. The arc length of the graph of $y = \dfrac{x^3}{6} + \dfrac{1}{2x}$ on the interval $\left[\dfrac{1}{2}, 2\right]$ is:

(a) $\dfrac{33}{16}$

(b) $\dfrac{27}{16}$

(c) $\dfrac{29}{16}$

(d) $\dfrac{35}{16}$

(e) $\dfrac{31}{16}$

14. $\int_0^{\pi/2} \sin 8x \, \cos 7x \, dx =$

(a) $\frac{3}{5}$

(b) $\frac{2}{3}$

(c) $-\frac{8}{15}$

(d) $-\frac{3}{15}$

(e) $\frac{8}{15}$

15. $\int \arctan x \, dx =$

(a) $x \arctan x + \frac{1}{2} \ln(1 + x^2) + c$

(b) $x \arctan x - \frac{1}{2} \ln(x^2) + c$

(c) $x \arctan x - \frac{1}{2} \ln(1 + x^2) + c$

(d) $x \arctan x - \frac{1}{2} \ln(1 + x) + c$

(e) $x^2 \arctan x - \frac{1}{2} \ln(1 + x^2) + c$

16. The volume of the solid generated by revolving the region bounded by the graphs of the equations $y = x$, $y = 3$ and $x = 0$ about the line $y = 4$ is

(a) 18π

(b) 16π

(c) 17π

(d) 20π

(e) 19π

17. $\int \dfrac{\sec^2 x}{\tan^2 x + 5\tan x + 6}\, dx =$

(a) $\ln\left|\dfrac{\tan x + 2}{\tan x - 3}\right| + c$

(b) $\ln\left|\dfrac{\tan x + 2}{\tan x + 3}\right| + c$

(c) $\ln\left|\dfrac{\tan x - 2}{\tan x + 3}\right| + c$

(d) $\ln\left|\dfrac{\tan x - 2}{\tan x - 3}\right| + c$

(e) $\dfrac{\tan x + 2}{\tan x + 3} + c$

18. If $\int \sin^3 x \, \cos^4 x \, dx = -\dfrac{1}{5}(\cos x)^m + \dfrac{1}{7}(\cos x)^n + c$, then $m + n =$

(a) 13

(b) 10

(c) 12

(d) 14

(e) 11

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$S = 2\pi \int_0^{\sqrt{2}} x\sqrt{1+4x^2}\,dx = \frac{\pi}{6}(27-1) = \frac{13\pi}{3}$.
2	**(c)**	$u = \tan 3x$: $\frac{1}{3}\int (u^3+u^5)du = \frac{u^4}{12} + \frac{u^6}{18}$, so $A+B = 30$.
3	**(e)**	Shells: $V = 2\pi \int_0^1 x(x^2+1)\,dx = \frac{3\pi}{2}$.
4	**(d)**	$u = \tan\frac{\theta}{2}$: integral becomes $\int_0^1 \frac{du}{1+u} = \ln 2$.
5	**(b)**	Limits -1 to 2; side $= -x^2+x+2$, area $= x^4 - 2x^3 - 3x^2 + 4x + 4$.
6	**(c)**	$\frac{6}{x} - \frac{1}{x+1} + \frac{9}{(x+1)^2}$, so $\ln\left\lvert\frac{x^6}{x+1}\right\rvert - \frac{9}{x+1} + c$.
7	**(c)**	$\frac{2}{e^{-x}+1} = \frac{2e^x}{1+e^x}$, so the value is $2\ln\frac{e+1}{2}$.
8	**(e)**	Complete the square, $u = x-2$: $2\arcsin\frac{x-2}{2} - \sqrt{4x-x^2} + c$.
9	**(c)**	$x = 5\sec\theta$: $\int \sec\theta\,d\theta$, giving $\ln\lvert x + \sqrt{x^2-25}\rvert + c$.
10	**(a)**	Parts, $u = \ln x$, $v = -\frac{1}{2x^2}$: $-\frac{\ln x}{2x^2} - \frac{1}{4x^2} + c$.
11	**(d)**	$6x = A(x^2+2x+4) + (Bx+C)(x-2)$ gives $A=1$, $B=-1$, $C=2$, so the sum is 2.
12	**(c)**	$t = \sqrt{x}$: $2\int t\sin t\,dt = -2t\cos t + 2\sin t$, so $a=-2$, $b=2$, $a+b=0$.
13	**(a)**	$1 + (y')^2 = (\frac{x^2}{2} + \frac{1}{2x^2})^2$; $L = [\frac{x^3}{6} - \frac{1}{2x}]_{1/2}^2 = \frac{33}{16}$.
14	**(e)**	$\sin 8x\cos 7x = \frac{1}{2}(\sin x + \sin 15x)$; integral $= \frac{1}{2}\cdot\frac{16}{15} = \frac{8}{15}$.
15	**(c)**	Parts with $u = \arctan x$: $x\arctan x - \int \frac{x}{1+x^2}dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + c$.
16	**(a)**	Washers: $V = \pi\int_0^3 [(4-x)^2 - 1]\,dx = 18\pi$.
17	**(b)**	$u = \tan x$: $\int \left(\frac{1}{u+2} - \frac{1}{u+3}\right)du = \ln\left\lvert\frac{\tan x+2}{\tan x+3}\right\rvert + c$.
18	**(c)**	$u = \cos x$: $-\frac{u^5}{5} + \frac{u^7}{7}$, so $m=5$, $n=7$, $m+n=12$.